

Incentives in Surplus Food Distribution for Smart Cities and Beyond: An Activity Aware Solution



Surja Sanyal, Sajal Mukhopadhyay, Fatos Xhafa, Vikash Kumar Singh, and Banhi Sanyal

Abstract Food surplus donation has emerged as a popular means to counter the food wastage issue that the world has been facing in recent years. The fact that this also addresses the food insecurity challenge is an added bonus. The matching of the surplus donors with the receivers needs the constant availability of willing volunteers to make this movement a success. An incentive module attracts volunteer participation and ensures their constant availability for the smooth movement of surplus food. While the existing incentive mechanism gets the attention of the volunteers, it fails to account for the degree of activity of the volunteers. This work modifies the incentive scheme to ensure that the activities of the volunteers are taken into consideration. The proposed mechanism manages to discourage inactivity with lower incentives and promote proactivity with higher incentives while disbursing the same total incentive amount taking all volunteers together.

S. Sanyal

Department of Computer Science and Engineering, Indian Institute of Technology Guwahati, Guwahati, Assam, India
e-mail: hi.surja06@gmail.com

S. Mukhopadhyay

Department of Computer Science and Engineering, National Institute of Technology Durgapur, Durgapur, West Bengal, India
e-mail: sajal@cse.nitdgp.ac.in

F. Xhafa (✉)

Department of Computer Science, Universitat Politècnica de Catalunya, Barcelona, Spain
e-mail: fatos@cs.upc.edu

V. Kumar Singh

School of Computer Science and Engineering, VIT-AP University, Amaravati, Andhra Pradesh, India
e-mail: vikash.singh@vitap.ac.in

B. Sanyal

School of Computer Engineering, Kalinga Institute of Industrial Technology, Bhubaneswar, Odisha, India
e-mail: banhisanyal9@gmail.com

Acronyms

AAIM	Activity Aware Incentive Mechanism.
FDRM-CA	Food Donor to Receiver Matching with Chronological Acceptance.
ICT	Information and Communications Technology.
IoT	Internet of Things.

Glossary

Activity based rewarding	The reward amount is directly proportional to the level of participation of the volunteer in surplus movement.
Chain of volunteers	The contiguous list formed by the volunteers present in the shortest path of the delivery person.
Delivery person	The volunteer who actually delivers a surplus food donation.
Donor-receiver relations	Informal relations that build up due to interactions for surplus movement.
Expected payment	The expected earnings of any arbitrary volunteer.
Exploit and explore	The concept of using the higher rewarding option more frequently, and using the unknown options less frequently only to find a more rewarding option.
Exploitable volunteers	Those volunteers who are actively working and will be rewarded by the system more frequently so as to motivate them to participate more.
Exploitation	The more frequent utilization of a highly rewarding option.
Exploration	The less frequent utilization of unknown options in search of an option that rewards higher than the current highest rewarding one.
Flat rewarding	The reward amount does not consider the level of participation of the volunteer in surplus movement.
Food habits	The practices of an individual that relate to the production, handling, and consumption of food.
Food supply chain	The processes responsible for bringing food from the fields to our plate.
Food insecurity	Inability to get regular access to enough safe and nutritious food for normal growth and development.
Food wastage	Food that is consumable but is consciously discarded.

Hyperparameter	The metadata variable of a mechanism which has a direct effect on the behaviour of the mechanism.
Identity duplication	Using multiple identities to obtain multiple resources/services.
Inactivity penalty	The less incentive that a volunteer gets when the volunteer does not participate actively in surplus movement.
Incentive mechanism	The mechanism that runs under the hood of the incentive module.
Incentive module	The module of the system that calculates and disburses incentives to the volunteers.
Invitations	Volunteers inviting other probable volunteers to the system.
Matching module	The module of the system that matches donors/donations with receivers/receipt requests through volunteers.
Mobility of the surplus	The pickup and delivery of the surplus food from its donor to its receiver.
Multi-armed bandit mechanism	The mechanism that maximizes rewards by utilizing the currently highest rewarding option more frequently, and exploring the unknown options with less frequency in search of a more rewarding option.
Operational practices	The methods that are used to operate a service/business.
Operational identity	The identity of a volunteer used to register and execute the surplus food donation requests.
Parent-children construct	The one to many hierarchy formed by volunteers consisting of one inviter (parent) and many invitee (children) volunteers.
Passive income	Income received by a volunteer generated by the work of other volunteers.
Perishable food items	Food items that spoil quickly and need immediate refrigeration.
Proactivity bonus	The extra incentive that a volunteer gets when the volunteer participates actively in surplus movement.
Probabilistic selection	The selection of volunteers based on probability/chance.
Scalable	Capable of change, often increase, in size.
Shift of funds	The increase or decrease of individual volunteer incentive amounts.
Shortest path	The route from the delivery person to the volunteer directly attached to the system without visiting any volunteer twice.

Smart cities	A municipality that uses Information and Communications Technology (ICT) for increased efficiency, information exchange, and public and government welfare.
Stabilize	Dampening of the fluctuations in the availability of volunteers.
Surplus food distribution	The redistribution of the extra food that was prepared, but is not being consumed.
Surplus food	The excess food that was prepared, but is not being consumed.
Sustainability	The ability to stay afloat in the long run.
Sybil attacks	To obtain multiple resources/services by operating from multiple identities so as to starve others from their rightful resources/services.
Technological platforms	The Information and Communications Technology (ICT) and Internet of Things (IoT) based platforms.
Top performers	Those volunteers whose participation in surplus distribution is more than some predefined threshold level.
Total incentive disbursed	The cumulative incentive disbursed by the system for one successful surplus delivery.
Volunteer participation	The measure of active surplus distribution by volunteers.
Volunteer hierarchical structure	The tree like structure formed by new volunteers joining the system through the invitation of an existing volunteer.

1 Introduction

Two major challenges that our society faces are food wastage and food insecurity. Surplus food distribution has surfaced as a solution that can counter both these issues at once. With the increase in the numbers of existing and upcoming smart cities, this solution has become more scalable. These smart cities equip the movement with the use of Information and Communications Technology (ICT) and the Internet of Things (IoT). These provide excellent platforms that make the communication between the parties faster and enable the surplus to reach to a broader audience. Most importantly, the ICT and IoT based platforms provide an infrastructure that helps in the proper traceability and accountability of the surplus donations. This helps in providing greater transparency to the movement and better satisfaction to the involved parties.

In our previous work, we have provided an inaugural algorithm, Food Donor to Receiver Matching with Chronological Acceptance (FDRM-CA), that pairs donors

with receivers of surplus food via willing volunteers [16]. It is noted that the availability of the volunteers fluctuates a lot and the success of the surplus movement empirically relies on their availability. Therefore, we have also provided an incentive module that goes hand in hand with the matching module stated above to boost and stabilize the availability of the volunteers who play a major role in the mobility of the surplus food [17].

While the existing incentive mechanism does attract more participation from the volunteers in addition to stabilizing their availability, it does not take into account the proactive behaviour of volunteers. It awards active and inactive volunteers alike, which may not be very encouraging for those who contribute proactively. In addition, volunteers may be encouraged to create an operational identity to invite probable proactive volunteers while they themselves can be passive participants only cashing in their share of the rewards. In this work, we modify the aforementioned incentive module so that it draws more participation from the volunteers and provides better incentives to the highly active participants while encouraging inactive participants to increase their contribution to the system. Our proposed mechanism achieves all the above mentioned enhancements while incurring the same cost to the incentive disbursal system.

Following is the organization of the paper. Section 2 presents the literature review in this field. Section 3 presents the problem formulation and the system structure along with the proposed mechanism. Section 4 provides the analysis of the same. Simulation results have been provided in Sect. 5 followed by Sect. 6 which presents the conclusion and the future scope for this work.

2 Related Work

Prior to this work, a foundation algorithm for matching food donors and receivers was provided [16], and an incentive mechanism was suggested for the volunteers who mediate this movement [17]. Herein, in this paper, we aim to provide an enhancement to the incentive mechanism so that the incentive system benefits further without incurring any extra volunteering cost overhead. That is to say, more active volunteer participation is encouraged by the modified incentive mechanism.

Food wastage is a major cause of concern for the whole world for the last few years hitting us hard on social and economic levels all around the globe [7]. Accumulating at an alarmingly high rate, the percentage of wasted food is now beyond 33% of the produce [18]. This itself amounts to billions of tons of food of which around a third is still perfectly edible but headed towards landfills [22]. There are several approaches to handle this serious concern. Amongst them, surplus food distribution has captured the fancy of the masses since it fights not only food wastage, but also food insecurity. Research is underway on the reinforcing effects of the Information and Communications Technology (ICT) on this long thriving notion. Design changes to the food supply chain have been forced by perishable food items [6]. With the application of the Internet of Things (IoT), routing and run time quality monitoring

of the surplus has become a trend of the movement [11]. A steady increment in global popularity is observed for the surplus distribution platforms that are based on ICT [21]. The movement affects daily chores of people in terms of their interests and practices. On the other hand, organisations and cultures also have an impact on the movement itself making the people interact over food related topics [15].

Meanwhile, surplus distribution stands as a viable solution to the global climate change brought about by wasting food that is avoidable on the first place [12]. Volunteer participation and ICT mediation render these initiatives highly cost effective. A food collection body can be a great augmentation to the surplus distribution move by directly or indirectly sourcing, via Government mediated channels, the surplus [3]. A central surplus collection and distribution body using food distribution bags is the most economically viable solution. Hand in hand, ICT and IoT make a great pair in reducing the wastage of food and raising the awareness of the same among the masses [9]. A database of the current operational practices and methods of the market aids in the improvement of donor-receiver relations and leads to better suggestions on food habits [20].

Research with *OLIO* (<https://olioex.com/>), a surplus distribution app, reveals that policymakers may get encouraged in promoting massive application of ICT in the food industry owing to the fresh non-commercial relationships that form while interacting for surplus movement [8]. The promotion of surplus movement heavily relies on such prevailing legislative policies of the land [4, 5]. ICT currently exists in most food sharing community operations in the form of passive centralized communication platforms. This renders a broad scope of ICT usage in greater complexities that can aid in faster communication between interested parties, reach a broader audience and achieve better timings in food delivery while clocking better metrics in terms of tracking and scalability. An incentive mechanism delivered over an ICT based platform further increases and stabilizes the availability of volunteers making the movement of the surplus smoother and reachable to a broader audience, contributing further to the sustainability of the movement [17].

3 Algorithms and Methodology

With the support of technological platforms available in the existing and upcoming smart cities, surplus distribution has become a bigger phenomenon than ever. The addition of the incentive module only accelerates the same by promoting and stabilizing the availability of volunteers who take the major charge in the forward march of the movement. However, Active volunteers look forward to a greater reward from the system than the passive ones. This is addressed and discussed in this section.

3.1 Notations and Problem Formulation

Before we take up the aforementioned issue, we present the following notations that will help our problem representation.

- λ : Incentive amount given to a delivery person (volunteer). Total incentive disbursed for all volunteers taken together will be twice of this amount.
- n : Total count of volunteers taking the shortest path from the delivery person to the incentive module.
- κ : Total count of volunteers selected for incentive disbursal. The delivery person, who is always rewarded, is not included here.
- δ : Maximum allowable count of volunteers in the aforementioned chain of n volunteers to whom incentives will be disbursed.
- l_E : Percentage of most active volunteers from the chain of n volunteers who will be exploited for further surplus deliveries.
- d_E : Percentage of the final selection of k volunteers consisting solely of the aforementioned exploitable volunteers.
- Γ : Sequential list of the aforementioned n volunteers starting with the delivery person as the 0^{th} volunteer.
- Δ : Sequential list of the activity counts of the n volunteers mentioned above.

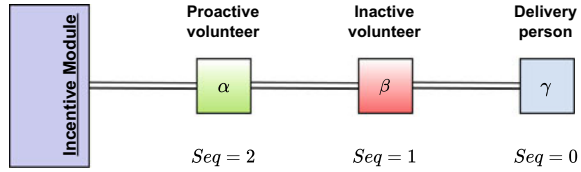
While the matching algorithm matches the donors with the receivers, the incentive module ensures a consistent required availability of volunteers to mobilize the surplus. However, not all enrolled volunteers remain active in the system or participate equally in the activity. Therefore, the incentive module must differentiate, in terms of incentives, between those who proactively participate in the movement and those who do not. With a higher reward for the proactive volunteers, the system can be sure to retain their willingness to contribute.

3.1.1 The Proposed Incentive Module

We present the proposed Activity Aware Incentive Mechanism (AAIM) inspired by the multi-armed bandit mechanism [1, 13, 19], which rewards proactivity and penalizes inactivity. We use the settings similar to the one used in our previous work [17] based on the incentive scheme put forward in [2, 10, 14]. A representative of the same is provided in Fig. 1. Here, the volunteer γ delivers the surplus and makes the whole chain of volunteers, that is, α , β and γ eligible for incentives. However, β is an inactive person here and receives passive income. Whereas, α is a proactive participant and expects the system to recognize the extra effort in terms of an increased incentive.

Although, the issue seems simple and an over the top solution may be to increase the incentive of the proactive volunteer by some factor to keep the person interested in contributing more to the system, the effects of the same would be undesirable. Consider the step where only $\kappa = \lfloor \sqrt{n} \rfloor$ volunteers are rewarded. This is done to

Fig. 1 Volunteer structure and proactivity for a successful delivery by volunteer γ



minimize the cost to the system. Therefore, increasing the incentive amounts of proactive volunteers would outright hurt the primary aim of cost minimization. On the other hand, the less active volunteers would definitely not be interested in contributing more if the system does not discriminate between the active and the passive volunteers. Thus, we propose a clever shift of funds from the passive volunteers to the active ones so as to keep the total cost to the system the same. This takes care of promoting the proactive nature of volunteers while encouraging the passive volunteers to be more active by penalizing them for a relatively lower activity count.

In Fig. 1, volunteer γ delivers the surplus. The volunteer hierarchical structure is formed as the volunteers invite other volunteers to the system who join through the inviter volunteer resulting in the parent-children construct. This parent-children construction of the volunteers is highlighted in Fig. 1. In this figure, for the surplus delivery, the chain of volunteers starts from the 0^{th} volunteer γ , who actually delivers the surplus food. The chain consists of all the volunteers who are in the shortest path from γ to the volunteer α who is directly connected with the system. The chain consists of the volunteers γ , β and α , labelled with sequence numbers $Seq = 0, 1$ and 2 , respectively. These are the only volunteers considered for the disbursement of incentives for this surplus delivery completed by γ . Volunteer α is a proactive member of the chain while volunteer β is not. If selected, both these volunteers receive incentives using the formula $\Delta_i = \$\frac{\lambda}{2^i}$ for a sequence number $Seq = i$. In the settings presented in Fig. 1, the existing mechanism will calculate incentives for γ as $\$\frac{\lambda}{2^0} = \λ , β as $\$\frac{\lambda}{2^1} = \$\frac{\lambda}{2}$, and α as $\$\frac{\lambda}{2^2} = \$\frac{\lambda}{4}$. The incentive amounts are directly proportional to the contribution of the respective volunteer in the movement of the surplus and reflects that γ has the maximum contribution in the same as the delivery person. Since, β has invited γ , β gets the next highest reward. Similarly, α gets the third highest reward for inviting β into the system. However, the existing mechanism does not select all n volunteers in the chain for reward. It randomly selects the κ volunteers to reward from the n volunteers in the chain. The proposed mechanism employs a different concept for selecting the κ volunteers for reward. Here, it utilizes the *exploit and explore* method from the concept of the multi-armed bandit [1, 13, 19], and modifies it to suit the settings of our requirements. In the multi-armed bandit concept of exploit and explore, the better performing resource is exploited more often than the one that performs less. To suit our requirements, we modify the same to further introduce a probabilistic selection of volunteers from the high performing group as shown in Fig. 2. Following are the steps for the selection:

1. Select the top $l_E\%$ volunteers with the highest activity. Here, $l_E\% < 50\%$ is a hyperparameter.

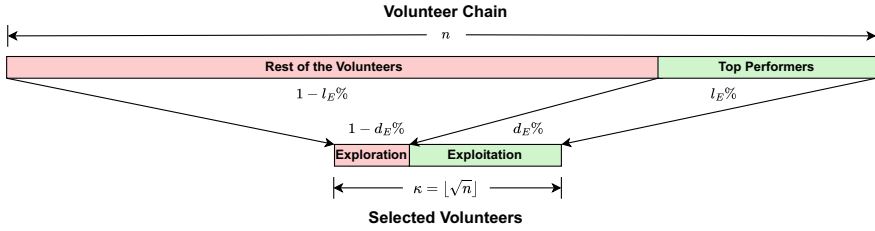


Fig. 2 Selection of $\kappa = \lfloor \sqrt{n} \rfloor$ volunteers from the chain of n eligible volunteers for incentive disbursement using the Exploit and Explore method

2. From the above selected list of volunteers, select at random $d_E\%$ of volunteers of the $\kappa = \lfloor \sqrt{n} \rfloor$ total volunteers to award incentives. Again, $d_E\% > 50\%$ is another hyperparameter.
3. Select the rest of the volunteers randomly from those volunteers having a lower activity to complete the selection of the κ volunteers.

Thus, we select the minority top contributors of the surplus mobility and exploit them further by selecting them more frequently (with higher probability) for providing incentives leading them to contribute more. Simultaneously, we select the less active volunteers less frequently for incentives, thereby effectively penalizing them for inactivity providing them with reason to increase their contribution. Note that the conditions $l_E\% < 50\%$ and $d_E\% > 50\%$ play an important role in the *exploit and explore* approach. It is this set of conditions that makes the exploitation part a majority as against the exploration part. Here, $l_E\%$ is the percentage of people who are the top performers of the volunteer team. We have taken $l_E\% < 50\%$, which indicates that we are flexible enough to fine tune it according to our need, and in the simulation we have varied it and have shown the impact. Similarly, $d_E\%$ dictates the percentage of people from the final selection who have been selected for exploitation from the aforementioned high performers. In our scheme, exploitation is the major percentage indicating that $d_E\% > 50\%$. We have also varied the same in the simulation and shown its impact. For the AAIM scheme, the incentive for a volunteer having the i^{th} sequence number is given by the following.

$$\Lambda_i = \begin{cases} \lambda/2^i, & \text{if selected,} \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

With the increased probability of the high performers for being selected for receiving incentives, the expected total incentives received by these volunteers become more than that of them who were inactive. This way, proactivity is rewarded at the cost of penalizing inactivity. We now prove the financial feasibility of the AAIM scheme by providing an upper limit for the total incentive disbursed by the scheme for one surplus delivery.

$$\Lambda_{total} = \sum_{i=0}^{\kappa=\lfloor\sqrt{n}\rfloor} \frac{\lambda}{2^i} < \sum_{i=0}^{\infty} \frac{\lambda}{2^i} = \frac{\lambda}{1-\frac{1}{2}} = 2\lambda. \quad (2)$$

In the AAIM scheme, sybil attacks (multiple identities of a single volunteer to reap higher incentives) are possible. To counteract the same to a certain extent we adopt to limit the maximum number of volunteers who can receive incentives for a single surplus delivery to δ . This will motivate volunteers to keep identity duplication at a minimum so as to remain eligible for further incentives. Thus, in contrast to the existing mechanism, the proposed AAIM mechanism promotes proactivity and demotes inactivity while incurring no extra overall cost for the same.

3.2 Proposed Mechanism

The inclusion of an incentive module increases and stabilizes the availability of volunteers for the movement of surplus food. The AAIM scheme will act along with the matching scheme to encourage proactive behaviour of volunteers and discourage inactivity. Algorithm 1 presents the working of the scheme. It starts with initializing the incentives of all n volunteers in the volunteer chain of the delivery person. The number of volunteers to reward is calculated as $\kappa \leftarrow \min(\delta - 1, \lfloor\sqrt{n}\rfloor)$. It then divides the n volunteers into two groups. One is the *Exploit* group having $l_E\%$ of the topmost contributors in the mechanism. Random selection from this group contributes to $d_E\% \times \kappa$ of the selected volunteers. The other group is the rest of the volunteers from the n volunteers whose contribution did not help them qualify for the first *Exploit* group. This is the *Explore* group from which random selection makes up the rest of the κ selected volunteers. In addition, the delivery person (0^{th} volunteer) always gets selected. Incentives are calculated for these $\kappa + 1$ people making that of the rest of the $n - \kappa - 1$ volunteers zero. This way, it connects with the multi-armed bandit mechanism where the best arms are selected for exploitation and from the rest exploration is performed. In this paper, the best arms signify the volunteers who are very active.

We start our incentive calculation from the volunteer who delivers the surplus food, and we denote this volunteer as the 0^{th} volunteer. This is denoted like that, as the incentive formula (1) will be in conformity with the incentives we are paying to the volunteers. With this logic the 0^{th} volunteer will get $\frac{\lambda}{2^0} = \lambda$ amount.

Algorithm 1: AAIM($\lambda, \delta, n, \Gamma, \Delta$)

```

1  $\Lambda \leftarrow \Phi$  // Initiate incentives.
2  $\kappa \leftarrow \min(\delta - 1, \lfloor \sqrt{n} \rfloor)$  // Count volunteers to reward.
3  $\Delta' \leftarrow \text{decreasing\_sort}(\Delta)$  // Sort by decreasing activity.
4  $\Delta'_{Cutoff} \leftarrow \Delta'[E\% \times n]$  // Get activity cutoff.
5  $V_{Exploit} \leftarrow \{t \mid (t \in \Gamma - \Gamma[0]) \cap \Delta[t] \geq \Delta'_{Cutoff}\}$  // Exploitation.
6  $V_{Explore} \leftarrow \Gamma - \Gamma[0] - V_{Exploit}$  // Exploration.
7  $\Gamma' \leftarrow \Gamma[0] \cup \text{random}(V_{Exploit}, d_E\% \times \kappa)$ 
    $\cup \text{random}(V_{Explore}, (1 - d_E\%) \times \kappa)$  // Assemble volunteers.
8 for  $i \leftarrow 0$  to  $n - 1$  in  $\Gamma$  do // For each volunteer.
9   if  $\Gamma[i] \in \Gamma'$  then // If selected.
10     $\Lambda[i] = \lambda/2^i$  // Disburse incentive.
11   else
12     $\Lambda[i] = 0$  // No incentive.
13 return  $\Lambda$  // Return incentives.

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4 Algorithm Analysis

Lemma 1 *The expected payment of the j^{th} volunteer is bounded by $\frac{2\lambda}{n}[\mu v + (1 - \mu)(1 - v)]$.*

Proof It is to be noted that with probability μ , a volunteer will belong to the top category and with $(1 - \mu)$ probability it will not. Again, we have that the volunteer may be selected either from the top category or from the non-top category. The probability of getting selected from the top category is v and from the other category is $(1 - v)$. Now, the probability for the event A_1 , that a volunteer will be selected is $\mu \cdot v + (1 - \mu)(1 - v)$. It is interesting to note that out of a chain of n volunteers, the selected volunteer maybe at any position in the chain. So, the probability that the volunteer will be selected at any i^{th} position can be calculated as:

$$\Pr(A_1 \cap A_2) = \Pr(A_1) \cdot \Pr(A_2) \quad (3)$$

where, A_2 is the event that the volunteer is from the i^{th} position. It is to be noted that they are all independent giving $\Pr(A_2) = \frac{1}{n}$ as all are equally likely to be selected. We can write Eq. 3 as:

$$\Pr(A_1 \cap A_2) = \Pr(A_1) \cdot \Pr(A_2) = \frac{1}{n}[\mu v + (1 - \mu)(1 - v)]. \quad (4)$$

With this probability we can calculate the expected earnings of any arbitrary volunteer j as:

$$\begin{aligned}
E[X] &= \sum_{i=1}^n \frac{\lambda}{2^i} \cdot \Pr(X = \text{the } i^{\text{th}} \text{ position for the selected case}). \\
&= \sum_{i=1}^n \frac{\lambda}{2^i} \cdot \frac{1}{n} [\mu v + (1 - \mu)(1 - v)] = \frac{[\mu v + (1 - \mu)(1 - v)]}{n} \cdot \lambda \cdot \sum_{i=1}^n \left(\frac{1}{2}\right)^i \\
&< \frac{[\mu v + (1 - \mu)(1 - v)]}{n} \cdot \lambda \cdot \sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^i = \frac{[\mu v + (1 - \mu)(1 - v)]}{n} \cdot \lambda \cdot \frac{1}{1 - \frac{1}{2}} \\
&= \frac{2\lambda}{n} [\mu v + (1 - \mu)(1 - v)]. \tag{5}
\end{aligned}$$

Observation 1 Plugging in the values of $\lambda = \$2000$, $n = 100$, $\mu = 0.25$ and $v = 0.75$ in Eq. 5, we get the expected maximum earning of an arbitrary volunteer as:

$$\begin{aligned}
E[X] &= \frac{2\lambda}{n} [\mu v + (1 - \mu)(1 - v)] \\
\Rightarrow E[X] &= \frac{2 * 2000}{100} [0.25 * 0.75 + (1 - 0.25)(1 - 0.75)] \\
\Rightarrow E[X] &= \$15.
\end{aligned}$$

5 Results

For the simulations, real data, sourced from OPEN DATA BARCELONA at <https://opendata-ajuntament.barcelona.cat/data/en/dataset/est-cadastre-carrecs-us>, was cast to the experiment settings using Python libraries. The dataset used for the simulation signifies cadastral positions according to the charge's use of the city of Barcelona. It has 22401 records. Each record states the district and the neighbourhood with their codes, the ownership description along with its valuation. The last column consisting of the value corresponding to the occupancy of land in some districts of Barcelona. So, in their data, each value corresponds to the valuation of a location. In our settings, we have used that value to correspond to a volunteer. We have cast the data like that way.

Volunteers were taken as 100 and successful surplus deliveries were varied in the range {2000, 4000, 6000, 8000, 10000}. Considering the number of volunteers available in a particular region, 100 is a sizable number. Similarly, the successful surplus deliveries were varied in the range {2000, 4000, 6000, 8000, 10000} as these are sizable and practical numbers. Our algorithm is scalable to experiment with higher numbers of volunteers and successful deliveries. The exploit group threshold was varied amongst $I_E\% = \{20, 40\}\%$. This was combined with the different values of selection of exploit group members for final incentive disbursal $d_E\% = \{60, 80\}\%$. The existing and the proposed methods were compared for the normalized incentives received by the volunteers when they are proactive versus when they are inactive.

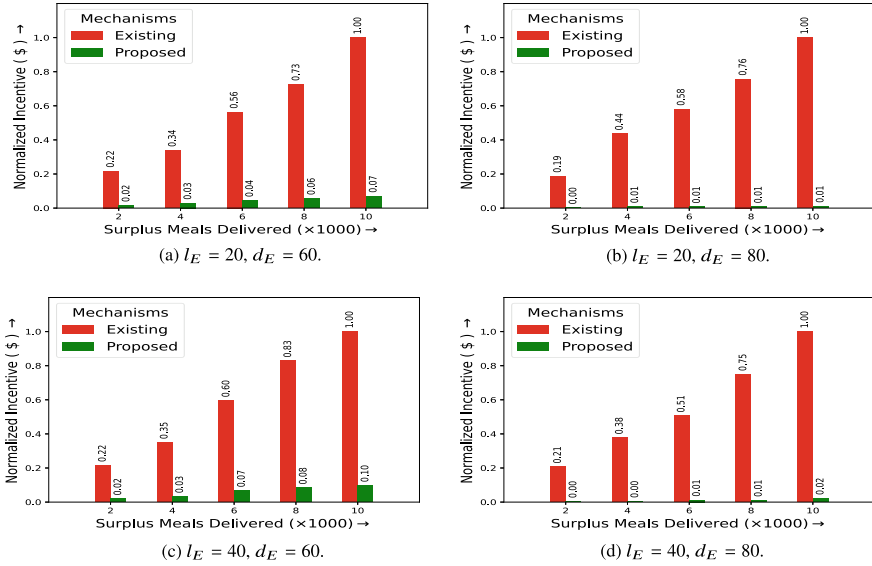


Fig. 3 The AAIM incentive scheme for *inactive* volunteers while varying $l_E\%$ and $d_E\%$

The total incentive disbursed by the AAIM method for these two types of volunteers were also compared with that of the existing method to deduce the proactivity bonus and the inactivity penalty amounts by the proposed mechanism.

5.1 Analysis

The results presented in Figs. 3, 4 and 5 are analysed here. Figure 3 shows the reduction in incentive of the inactive volunteers by the AAIM scheme when compared with the existing scheme for different values of exploitation threshold $l_E\%$ and selection from exploit group for incentive disbursal $d_E\%$. It shows that the proposed mechanism penalizes inactivity heavily. On the other hand, Fig. 4 shows the bonus that the proactive volunteers receive using the AAIM mechanism in contrast to that for the existing mechanism while varying $l_E\%$ and $d_E\%$. This shows that the proposed mechanism encourages the proactivity of the volunteers.

Lastly, Fig. 5 contrasts the activity based rewarding nature of the AAIM scheme against the flat rewarding nature of the existing scheme. For the inactive volunteers, the reduction in incentive by the proposed mechanism against the existing mechanism is equivalent to a penalty for inactivity. Similarly, the increased incentive disbursed by the proposed mechanism for the proactive volunteers when contrasted with the incentive rewarded by the existing mechanism is equivalent to a bonus for proactivity. Both these activity based penalty and bonus systems are enforced without incurring

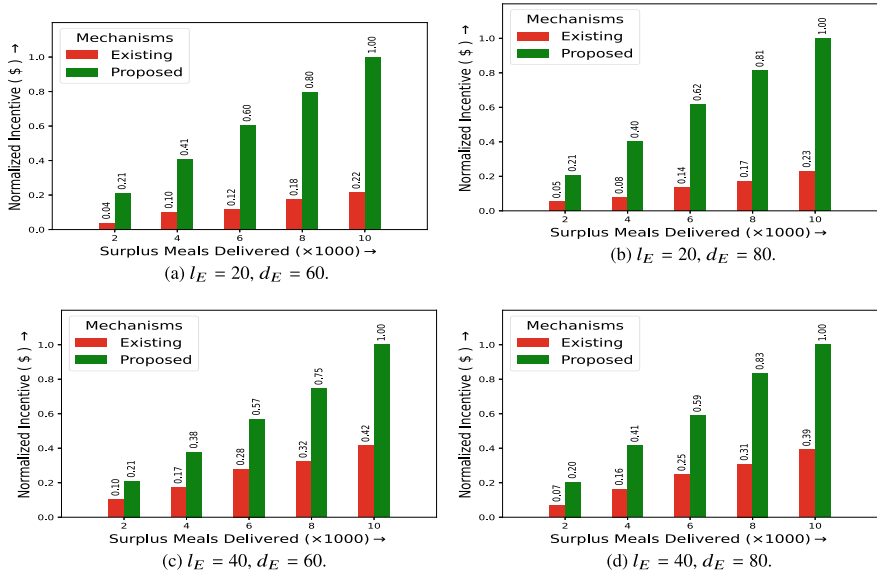


Fig. 4 The AAIM incentive scheme for *proactive* volunteers while varying $l_E\%$ and $d_E\%$

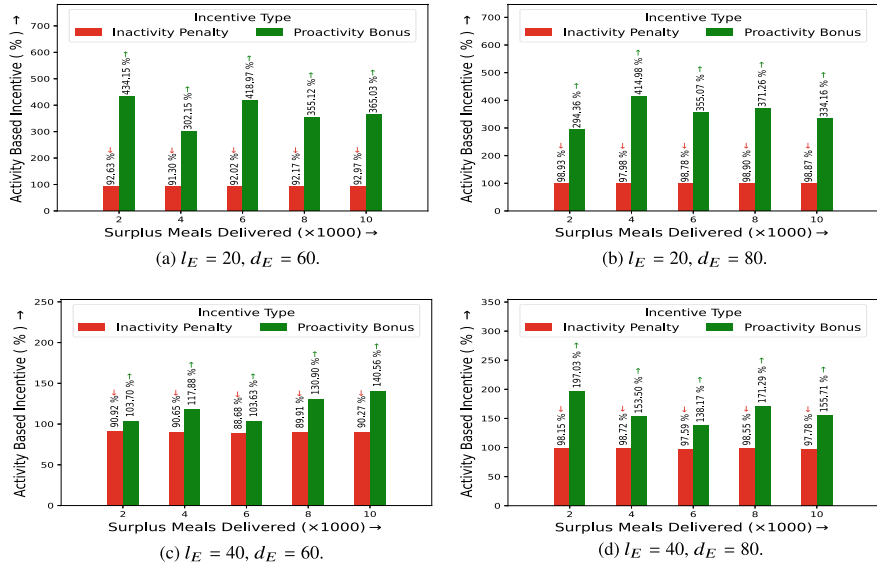


Fig. 5 The AAIM scheme bonuses and penalties for volunteers while varying $l_E\%$ and $d_E\%$

any extra cost to the incentive module, i.e., both the existing and the proposed schemes disburse the same total incentives when all volunteers are taken together.

6 Conclusion

Food wastage is a great challenge for the world, as is food insecurity. Surplus food distribution has come up as a great solution for both the challenges until deeper rectifications are able to transform the situation. Volunteer participation plays a pivotal role in the success of the surplus movement. Therefore, an incentive mechanism can not only stabilize their availability, but also encourage a greater participation. Existing incentive mechanisms provide for a basic start to this goal. In real life environments, all volunteers do not contribute equally to the movement and those who contribute more expect to be rewarded more for their proactive participation. However, the existing mechanisms do not account for the degree of contribution of the volunteers and reward all volunteers flatly with respect to the same.

The proposed incentive scheme uses the concept of the “multi-armed bandit problem” to take into account the degree of participation of the volunteers while considering for incentive disbursement. For inactive volunteers it reduces the incentive amount disbursed to them. This amount that was saved from the incentives for the inactive volunteers is transferred to the proactive volunteers. Thus, the proposed incentive mechanism discourages inactivity and promotes proactivity while incurring the same cumulative incentive amount disbursed when compared with that of the existing mechanism.

We note that the proposed incentive scheme is vulnerable to sybil attacks (identity duplication). Limiting the maximum possible number of volunteers to reward as a solution to the same may be counter productive towards encouraging the volunteers to join the movement through invitations. Designing and pruning a mechanism to address the sybil attacks is a challenging future work.

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